Introduction

Iron is an essential element for mammals, plants and some single celled organisms. It plays an important role in our bodies. It is found in a part of our blood called hemoglobin, allowing our blood to carry oxygen molecules from our lungs to the cells of our body. Our bodies cannot produce iron (hence, essential dietary element), but it is naturally present in many foods, including meats (beef, pork, turkey), produce (raisins, spinach, prunes) and nuts (walnuts, cashews, peanuts) and artificially added to many others (fortification).

In the 1920s, food fortification with vitamins and minerals started becoming commonplace in the U.S. and Europe to address several public health issues. Between the 1920's and 1950's, salt was fortified with iodine to prevent goiter, milk was fortified with vitamin D to prevent rickets, and flour (and bread) was fortified vitamin B and iron to prevent beriberi[1]. Since then, a lot of foods have been fortified with vitamins and minerals to prevent more vitamin and mineral deficiencies and related health issues and it is not uncommon to see a list of vitamins and minerals on food labels. Climate change has also increased the efforts to fortify foods, with research showing that increased atmospheric CO_2 levels lower the nutritional quality of foods, especially the iron and zinc content[2]. To make up for this lowered quality, food manufacturers around the world add iron to staple grains, such as fortified breakfast cereals.

The purpose of this experiment is to determine the amount of iron in a serving of cereal and compare your experimental result to values listed on the nutrition information of the cereal box. You will also compare your results with your classmates' results and determine which of the cereal brands contains the most iron.

First, you will extract the iron from a sample of cereal using a magnet. Next, you will dissolve the iron in a solution of hydrochloric acid (HCl) and hydrogen peroxide (H_2O_2). HCl will oxidize the iron (Fe(s)) to the Fe^{2+} oxidation state (Reaction CI1). The strong oxidizer, H_2O_2, will further oxidize the iron to Fe^{3+} (Reaction CI2). Next, you will add potassium thiocyanate (KSCN) to a diluted sample of the solution. Fe^{3+} and SCN^- will react (Reaction CI3) to produce a colored complex, $[Fe(SCN)_6]^{3-}$aq).

Note the molar ratio of $[Fe(SCN)_6]^{3-}$ to Fe^{3+} to Fe^{2+} to Fe(s) for your calculations.

$$Fe(s) + 2\,HCl(aq) \longrightarrow Fe^{2+}(aq) + 2\,Cl^-(aq) \longrightarrow H_2(g) \qquad \text{(Reaction CI.1)}$$

$$2\,Fe^{2+}(aq) + H_2O_2(aq) \longrightarrow 2\,Fe^{3+}(aq) + 2\,OH^-(aq) \qquad \text{(Reaction CI.2)}$$

$$\underset{\text{pale yellow}}{Fe^{3+}(aq)} + 6\,SCN^-(aq) \longrightarrow \underset{\text{red−orange}}{Fe(SCN)6(aq)^{3-}} \qquad \text{(Reaction CI.3)}$$

The pale-yellow solution of Fe^{3+} and colorless thiocyanate ion (SCN^-) react to produce a red-orange colored species, $[Fe(SCN)_6]^{3-}$, which can be analyzed by visible spectroscopy.

The absorbance of $[Fe(SCN)_6]^{3-}$ gradually decreases over time, which can affect your results, so wait until you are ready to analyze your solution(s) before adding KSCN to them.

You know from the previous experiment that the concentration of a colored species is proportional to its absorbance as measured by a spectrophotometer according to Beer-Lambert law. To find the concentration of $[Fe(SCN)_6]^{3-}$ in your cereal samples, you must first prepare a calibration plot using standard solutions of $[Fe(SCN)_6]^{3-}$. Using the concentration of $[Fe(SCN)_6]^{3-}$ in your cereal samples, the molar ratios from the chemical reactions, and the dilution factors, you will then calculate the amount (in mg) of iron in your cereal sample.

[1]Bishai, D.; Nalubola, R. The History of Food Fortification in the United States: Its Relevance for Current Fortification Efforts in Developing Countries. *Economic Development and Cultural Change* **2002**, *51* (1), 37–53. https://doi.org/10.1086/345361.

[2]Myers, S. S.; Zanobetti, A.; Kloog, I.; Huybers, P.; Leakey, A. D. B.; Bloom, A. J.; Carlisle, E.; Dietterich, L. H.; Fitzgerald, G.; Hasegawa, T.; Holbrook, N. M.; Nelson, R. L.; Ottman, M. J.; Raboy, V.; Sakai, H.; Sartor, K. A.; Schwartz, J.; Seneweera, S.; Tausz, M.; Usui, Y. Increasing CO2 Threatens Human Nutrition. *Nature* **2014**, *510* (7503), 139–142. https://doi.org/10.1038/nature13179.

Learning Objectives

Extract iron from cereal with a magnet

Use oxidation-reduction (redox) reactions to oxidize Fe(s) to Fe^{3+}(aq)

Use Beer's law to determine the concentration of iron in a cereal sample

Compare class results to determine which cereal contains the most iron per serving size

Procedure

Work in groups of 2 students (pairs). You and your lab partners must share the tasks. Your TA will assign your group one of three cereals available for analysis.

Safety Precautions

Wear your goggles at all times in the laboratory.

- Use caution when working with hydrochloric acid (HCl, 6 M). If you spill acid on your skin or clothing, alert your TA and rinse the area promptly with plenty of water.
- If you leave the lab, take you gloves off and throw them away so you do not transport any hazardous materials that might be on the gloves outside of the lab. Get new gloves when you return to lab.
- The dissolving of iron step must be performed under a snorkel hood.

Equipment and Reagents

Our tap water contains iron, which will alter your results. It is important that all glassware is clean. **Rinse all glassware (beakers, test tubes, burets, etc.) thoroughly with deionized water before use. Condition pipets and burets with the solution being dispensed prior to use.**

Your Equipment Drawer

- Spatula
- Two 100-mL beakers (clean and dry)
- One stir bar
- 50-mL graduated cylinder

Table Drawers and Bench Top

- Plastic wash bottle
- Stirrer/hot plate

Common Equipment Areas

- Wide mouth jar (upper cabinet)
- Ziploc bag
- Plastic hexagonal dish (not cupcake cup)
- 5 g of assigned cereal

Reagents

- 40 mL 2 M hydrochloric acid, HCl
- 10 mL 3% hydrogen peroxide, H_2O_2

PART A - Collect Iron from Cereal Sample

1. Weigh ~5 g of cereal and place in a Ziploc bag. Use your fist or a wide mouth jar to crush the cereal to a fine powder (some larger pieces are acceptable) as shown in Figure CI.1.

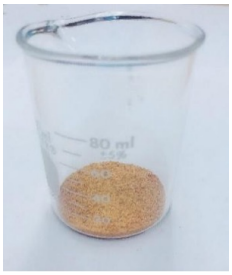


Figure CI.1: Crushed Cereal Sample

2. Tare a 100-mL clean and dry beaker and add about 3.0 g of the crushed cereal. Record the mass to 3 decimal places.

> Note that the procedure states: "about 3.0 grams" and not "about 3 grams". The extra zero (significant figure) in "3.0 grams" indicates that your measurement should be within the range of 2.95 to 3.04 grams, i.e. values that round to 3.0 grams (with 2 significant figures). Stating "about 3 g" would usually indicate a range between 2.50-3.49 grams, however not all values within this range would round to 3.0 grams with 2 significant figures. For example, 3.36 would round to 3 grams for 1 significant figure and 3.4 grams for two significant figures, making it within range for "about 3 grams," but not within range for "about 3.0 grams."

3. Add about 50 mL of deionized water to the beaker – enough to "cover" and saturate the cereal. Add additional water, up to 10 mL, if your cereal sample is not completely covered and saturated with the water.

4. Add a clean stir bar to the beaker.

5. Set up a stir plate and place the sample beaker on the stir plate. Set the heat to 113" and stir at medium speed ("5" or "6") for 30 minutes. Adjust stir speed as needed to achieve good movement of the sample solution without stirring excessively fast.

> The purpose of this process is to rehydrate the cereal and make the iron extraction more efficient. Warming up the cereal in warm water hydrates the dry cereal and creates a mushy, mostly liquid mixture and makes it easier to separate the iron particles from the cereal.

6. **While your cereal heats up, begin preparing and analyzing your $[Fe(SCN)_6]^{3-}$ standard solutions.**

After the First 30 Minutes of Stirring

Caution

You are about to use a concentrated acid. If you come into contact with the acid, flush the area with water from the sink for 5–10 minutes. Notify your TA immediately.

7. To a clean and dry 100-mL beaker, add 40 mL of 2 M HCl and 10 mL of 3% H_2O_2. The exact volume is not critical, use marks on beaker to get close to 40 mL of HCl and then add H_2O_2 to bring the volume to the 50 mL mark. Place the solution under the snorkel hood.

8. Remove the cereal beaker from the stir plate after a total of 30 minutes of stirring and temporarily place it on a heat resistant pad.

9. Carefully remove the stir bar with a spatula and place in a plastic weigh boat. You should see iron particles on the stir bar. See Figure CI.2.



Figure CI.2: Stir Bar with Iron

10. Gently wash the stir bar, using only a few of drops of deionized water, to remove the cereal from the stir bar. Be careful not to lose any iron while rinsing.

11. Carefully transfer the washed stir bar to the beaker containing HCl/H_2O_2.

12. Place the HCl/H_2O_2 beaker on the stir plate and stir the solution to a setting of "4" until all iron dissolves.

13. After stirring, remove the stir bar. Rinse the stir bar with deionized water, with the rinse going into the HCl/H_2O_2 beaker.

14. Place the rinsed stir bar back into the cereal beaker and stir for another 30 minutes with heating. The stir bar will collect any remaining iron.

15. After 30 minutes, remove the second stir bar, wash gently as before, and add to your beaker of HCl and H_2O_2, and stir again.

PART B - Preparation and Analysis of $[Fe(SCN)_6]^{3-}$ Standard Solutions

Our tap water contains iron, which will interfere with your results, so rinse everything thoroughly with deionized water. Prior to use, condition the burets with the appropriate solution..

Equipment

Your Equipment Drawer

- Two 100-mL beakers (clean and dry)
- 250-mL beaker (clean and dry)
- Six 6 inch test tubes

Table Drawers and Bench Top

- Wooden test tube support

Common Equipment Areas

- Two 25-mL burets
- 50-mL buret

Reagent Hood

- 30 mL 0.00020 M iron(III) chloride, $FeCl_3$
- 60 mL 1.0 M potassium thiocyanate, KSCN
- 80 mL 2 M hydrochloric acid, HCl

1. Label the burets and test tubes so you don't confuse the samples or solutions.

Caution

You are about to use a concentrated acid. If you come into contact with the acid, flush the area with water from the sink for 5–10 minutes. Notify your TA immediately.

2. Rinse and fill burets with the solutions that you will use for preparation of solutions (0.00020 M $FeCl_3$ and 1.0 M KSCN in the 25 mL burets and 2 M HCl in the 50 mL buret). Record the concentrations of the solutions (from the reagent bottles) used to prepare the standard solutions in your lab notes.

3. Table CI.1 describes the amount of 0.00020 M $FeCl_3$ and 2 M HCl needed to prepare the blank solution and five standard solutions of $[Fe(SCN)_6]^{3-}$. **Do not add the KSCN solution until you are ready to make the absorbance measurements. Once the KSCN solution is added, the sample is only stable for about 15–20 minutes.**

Table CI.1: Volumes required to prepare standard solutions

Solution	0.000200 M $FeCl_3$ (mL)	2.00 M HCl (mL)	1.00 M KSCN (mL)
Blank	0	10.00	5.00
1	0.50	9.50	5.00
2	1.00	9.00	5.00
3	2.00	8.00	5.00
4	4.00	6.00	5.00
5	8.00	2.00	5.00

4. Add 10.00 mL of 2 M HCl to the "blank" test tube.

5. Prepare each standard solution by first adding the prescribed volumes of $FeCl_3$ solution and HCl solution to your labeled test tubes.

6. Record the exact volume of each solution dispensed to 2 decimal places. Each test tube will have a total volume of 10.00 mL at this point.

> Note: Save these labeled burets! You will need to create a separate solution later to determine wavelength of maximum absorbance.

PART C - Dissolve the Collected Iron and Dilute Iron Solution

Equipment

Your Equipment Drawer

- Long glass stir rod
- 250-mL beaker
- 6 inch test tube

Table Drawers and Bench Top

- Pipet bulb

Common Equipment Areas

- 25-mL pipet
- 10-mL pipet
- Two 100-mL volumetric flasks
- Parafilm®

1. With the stir bar in the HCl/H_2O_2 beaker, stir at medium speed until all the iron on the stir bar dissolves. Your solution should be faint yellow color as seen in Figure CI.3.



Figure CI.3: The iron solution

2. Place a stir rod at the mouth of the beaker and slowly pour the resulting iron solution into a 100-mL volumetric flask. Do this carefully so you do not lose any of the sample solution or pour out the stir bar.

3. Use your wash bottle to rinse the beaker and stir bar with about 10 mL of deionized water then pour into the 100-mL volumetric flask. Rinse two more times, adding the rinse to the 100-mL volumetric flask each time.

> Note: Be sure to rinse the sides of your 100-mL beaker to ensure you get all of the iron.

4. Add deionized water to the volumetric flask to bring the solution volume to the 100.00 mL mark.

5. Seal the opening of the flask with about a quarter of a square (1/4 square) of Parafilm (See Figure CI.4) and invert several times to mix. While inverting, you should shake the flask for the liquid to allow more movement to ensure your solution is at equilibrium.

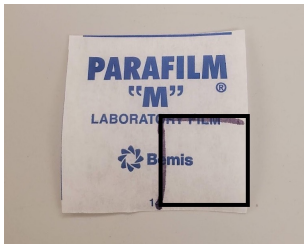


Figure CI.4: A quarter of a square of Parafilm.

6. Use a 25.00-mL volumetric pipette to transfer exactly 25.00 mL of the solution into a clean 100-mL volumetric flask and dilute to mark with deionized water.

7. Transfer your diluted iron sample to a clean, dry 250-mL beaker. Use a clean 10.00-mL pipette to transfer exactly 10.00 mL of the diluted sample solution to a test tube. Label the test tube.

PART D - Data Collection

Equipment

Your Equipment Drawer

- 6 inch test tube

Common Equipment Areas

- Parafilm®
- Wireless spectrophotometer
- Plastic Cuvette

1. Connect the SpectroVis Plus spectrophotometer to your iPad.

2. Open the Spectral Analysis app on the iPad.

3. Press the button on the front of the spectrometer to turn it on.

4. Tap **Connect a Spectrometer** then find your spectrometer by matching the ID# on your instrument with those displayed. Tap **Connect** then **DONE**.

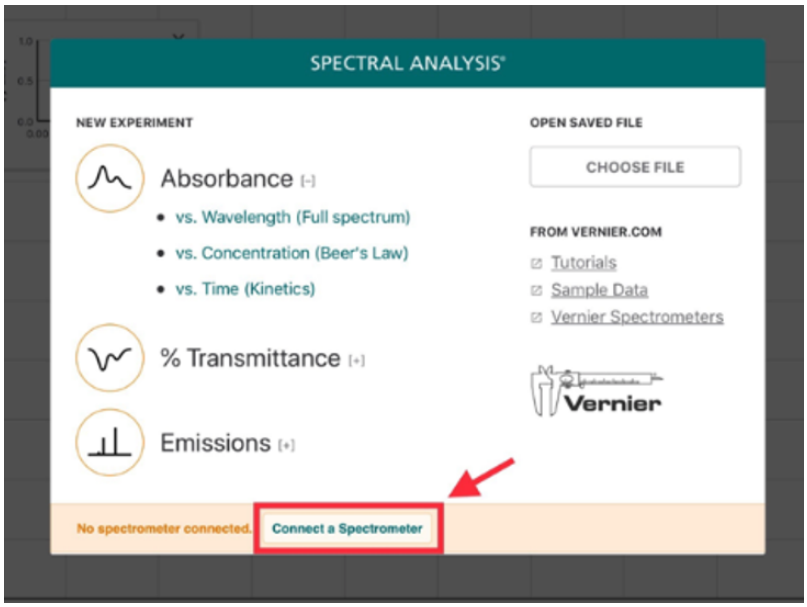


Figure CI.5: Connect a spectrometer

Calibrate the Spectrometer and Determine the Wavelength of Maximum Absorbance

5. Select **vs. Concentration (Beer's Law)**. See the red rectangle in Figure CI.6.

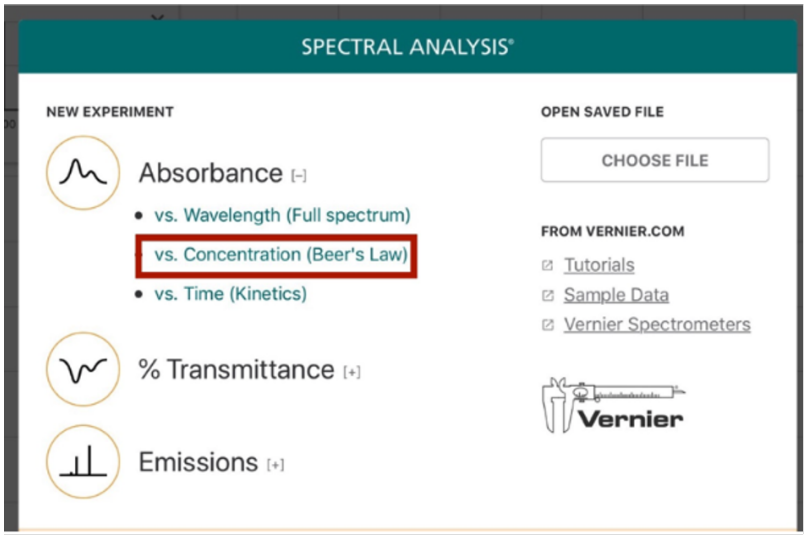


Figure CI.6: Spectral Analysis Absorbance vs Concentration

6. Allow the device to complete warm up.

7. From the buret, add 5.00 mL of 1.0 M KSCN to the "blank." Cover with Parafilm and invert to mix.

8. Rinse and fill the cuvette about 3/4 full with the "blank" solution. Wipe the outside of the cuvette with a KimWipe.

9. When prompted, place the cuvette containing the "blank" solution into the spectrometer and select **Finish Calibration**.

Choose the Wavelength of Maximum Absorbance

10. Prepare a separate standard solution containing $[Fe(SCN)_6]^{3-}$ by combining 3.00 mL 0.00020 M $FeCl_3$, 7.00 mL 2 M HCl, and 5.00 mL 1.0 M KSCN in a 6 inch test tube. You will use this solution only for the selection of the wavelength of maximum absorbance; **it is separate from your standards 1-5 solutions you prepared earlier**.

11. Use a small piece (1/4 of a square) of Parafilm to seal the test tube and invert to mix.

12. Rinse and fill a cuvette about 3/4 full with the standard solution. Wipe the outside of the cuvette with a KimWipe.

13. Insert the cuvette containing the standard solution into the spectrometer.

14. Wait for about 10 seconds, or until the spectrum resolves.

15. Tap the spectra to examine the curve. Find the wavelength at which the absorbance is a maximum (λ_{max}) by moving the cursor to the highest point of the peak absorbance. See Figure CI.7 below. The wavelength will appear in the Selected Wavelength box.

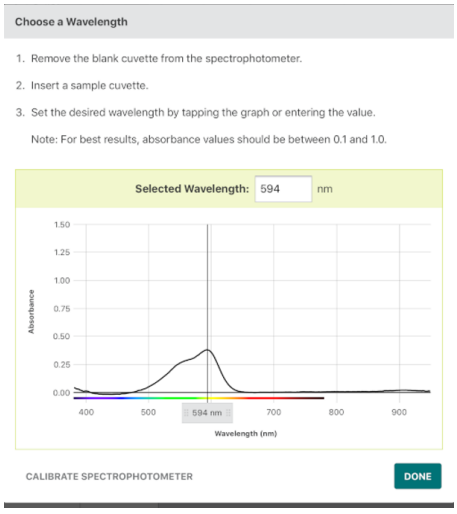


Figure CI.7: Choose the Wavelength of Maximum Absorbance

16. Take a screenshot of the graph and selected wavelength.

17. Click **Done** once you have selected your wavelength.

Generate a Calibration Plot using the Standard Solutions

18. From a buret, add exactly 5.00 mL of 1.0 M KSCN to each test tube. Seal the tops of the test tubes tightly with small pieces of Parafilm and invert 5-6 times to mix. You should now have 8 test tubes: your sample, the blank, the solution you prepared to determine the wavelength of maximum absorbance, and 5 standard solutions. Each test tube will have a total volume of 15.00 mL.

    > Your standard solutions should have the red-orange color. (See Reaction CI3 in the Introduction section). Arrange your five standard solution test tubes in order of concentration. You should see that the red-orange color becomes more intense as the iron concentration increases.

19. When you are ready to analyze the standard solutions press **COLLECT**.

20. Rinse the cuvette with a small amount of the standard solution #1 by rolling the solution in the cuvette.

21. Fill the cuvette about 3/4 full with the standard solution.

22. Wipe the outside of the cuvette with a KimWipe then insert it into the spectrophotometer.

23. Wait about 10 seconds for the absorbance value to stabilize then tap **KEEP**.

24. Enter the concentration of that standard in mol/L and tap **Keep Point**.

    > NOTE: 1.00×10^{-3} must be entered as 1.00e-3

25. Repeat steps with the remaining standards. Record the absorbance values in the table below.

26. Once all data is collected (i.e. after you have measured the absorbance of all the standard solutions), tap **STOP**.

Calibration Plot for Standard Solution

27. Click the picture of the graph in the bottom-left of the screen and select **Apply Curve Fit**.

28. Select Linear and the tap **APPLY**. (See the red arrow and circle in Figure CI.8.)

29. Take a screenshot of your calibration plot. Record your line equation.

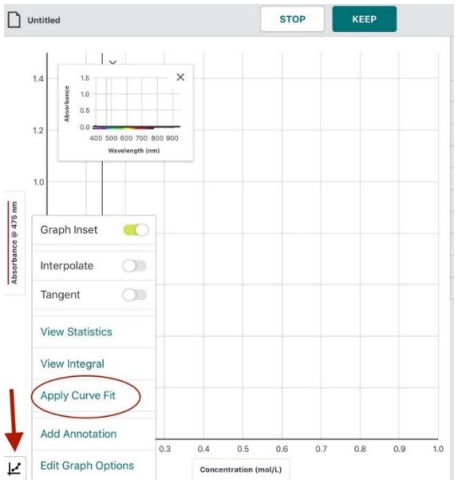


Figure CI.8: Calibration Curve

Measure the Absorbance of Cereal Sample

30. Make sure you add 5.00 mL of 1.0 M KSCN to the test tube of your cereal sample solution to have a total volume of 15.00 mL.

31. Rinse and fill the cuvette 3/4 full with your cereal sample solution and wipe the outside of the cuvette with a KimWipe.

32. Insert the cuvette into the spectrophotometer.

33. Wait about 10 seconds for the absorbance reading to stabilize.



Figure CI.9: The cereal sample solution ($[Fe(SCN)_6]^{3-}$)

Data from Other Cereals

To compare the amount of iron in cereal, you will be given data from two different cereals than the one your group worked with. You will use this data to calculate the grams of Fe in a serving size of the other types of cereal.

- Type of Cereal
- The mass of cereal (g) in a serving size
- The mass (mg) and % Daily Value (DV) of iron in a serving size
- The equation of the line for the calibration plot
- Mass Cereal used in Sample (g)
- The absorbance value of the cereal sample solution

WASTE DISPOSAL AND CLEANUP

- Rinse off the plastic hexagonal weighing dish with DI water and wipe it clean, then return the weighing dish to the reagent table so it can be used again.
- All solutions can be poured down the drain followed by a lot of running water.
- Rinse all glassware once with tap water followed by a minimum of two rinses with deionized water from your wash bottle.
- Return all reagents and equipment to the proper places.
- Wipe and dry the tops of the lab benches.
- Keep your splash goggles on until you are leaving lab.

Data Collection

PART A - Collect Iron from Cereal Sample

Type of Cereal: _____

Mass (g) of cereal in serving size: _____ (g)

Mass (mg) of iron in a serving size: _____ (mg)

Percent Daily Value (DV) of iron in a serving size: _____ %

Mass (in g) of crushed cereal: _____ (g)

PART B - Preparation and Analysis of $[Fe(SCN)_6]^{3-}$ Standard Solutions

Concentration of Iron (III) Chloride, $FeCl_3$: _____ (M)

Concentration of Potassium Thiocyanate, KSCN: _____ (M)

Concentration of Hydrochloric Acid, HCl: _____ (M)

Report Table CI.1: Actual Volumes Dispensed (to 2 decimal places) for Each Sample

Solution	0.00020 M $FeCl_3$ (mL)	2 M HCl (mL)	1.0 M KSCN (mL)
Blank			
1			
2			
3			
4			
5			

PART D - Data Collection

Wavelength of Maximum Absorbance (λ_{max}): _____

Report Table CI.2: Concentration and Absorbance of each Standard Solution

Standard	Concentration	Absorbance
1	_____	_____
2	_____	_____
3	_____	_____
4	_____	_____
5	_____	_____

Equation of Calibration Plot: _____

Absorbance of Cereal Sample Solution: _____

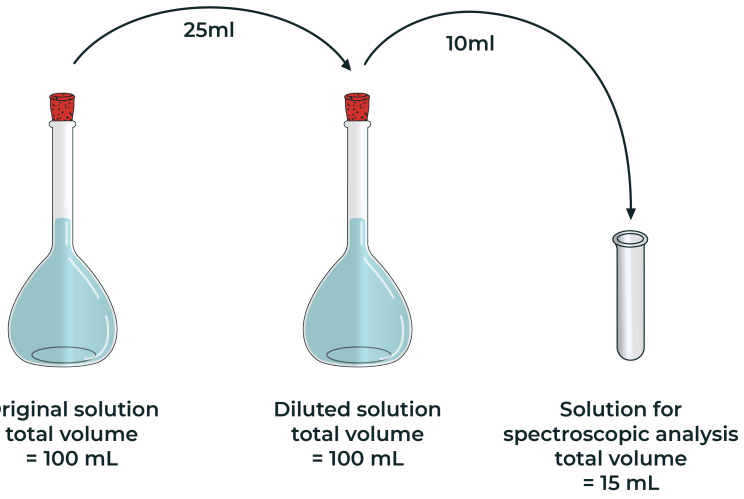
Report Table CI.3: Concentration and Absorbance of both Cereal Samples

	Cereal 1: _____ (cereal name)	**Cereal 2: _____ (cereal name)**
Mass (g) of Cereal used in Sample	_____	_____
Mass (g) of Cereal in a Serving Size	_____	_____
Mass (mg) and % Daily Value (DV) of iron in a serving size	_____	_____
Equation of line for calibration plot	_____	_____
Absorbance of the Cereal Sample	_____	_____

Report Form

Calculation #1. Using the absorbance value for your diluted cereal sample, calculate the concentration of Fe^{2+} in your **original** 100.00 mL solution.

You diluted your sample solution twice: when you transferred a measured volume from the 100 mL volumetric flask to another volumetric flask and again when you transferred a measured volume to a test tube and added KSCN. Use the concentration (molarity) of iron from above and the dilution factors to determine the concentration of iron in the original 100 mL volumetric flask.



Show calculation 1.

Calculation #2. Calculate the experimental mass of Fe(s) (mg) in your cereal sample using your experimental results. Use the concentration of $[Fe(SCN)_6]^{3-}$ in the original 100-mL volumetric flask to determine the mass of iron (in milligrams) in your original cereal **sample** (of about 3 grams).

Show calculation 2.

Calculation #3. Calculate the experimental mass of Fe(s) (mg) in a *serving size* of your cereal using your experimental results.

Refer to the information you recorded earlier for the mass of a serving size of the cereal you analyzed. Using the mass of iron in your cereal sample calculated above; determine the *experimental* mass (in mg) of iron in a **serving size**.

Show calculation 3.

Question #1. Compare the experimental mass of Fe(s) in a serving size to the actual value on the cereal box. Show a calculation to support your answer. (Hint: Which formula from Lab 1 is appropriate for this comparison?)

Question #2. Based on the class results, what conclusions can you draw about the iron content of these three types of cereal? Which type has the most iron according to class results? Which cereal has more iron according to the cereal labels? Refer to specific results in your explanation. (To answer this question, you will need to calculate and report the mass of iron in a serving size of the other types of cereal using the data provided by the other groups in your lab section.)

Question #3. What is the recommended amount of iron that should be consumed daily? Based on this number and your experimental results is the cereal you analyzed a good source of iron for the average adult? Why or why not? Cite your sources and refer to your data.